Math 110 — Week 5

Quotient maps, isomorphism theorems, dual spaces (Notes Ch. 9–10, LADR 3E–3F, HW5)

§1 Facts used from Weeks 1–4

(W1) An independent list extends to a basis. A linear map can be defined by choosing its values on a basis, and two linear maps that agree on a basis are equal (Weeks 2, 3).

(W2) Rank–nullity: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$. T is injective $\iff \operatorname{null} T = \{0\}$ (Week 3).

(W3) If $\dim V = \dim W < \infty$, then injective $\iff$ surjective $\iff$ isomorphism. Finite-dimensional spaces are isomorphic exactly when their dimensions are equal (Week 4).

(W4) $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W$. The matrix of T: column j holds the coordinates of Tv_j (Week 3).

(W5) Quotient space V/U: its elements are the sets $v+U$; $v+U = w+U \iff v-w \in U$; $(v+U)+(w+U) = (v+w)+U$, $\lambda(v+U) = \lambda v + U$. A formula that uses the name v must be checked to be *well-defined*: renaming the same set must not change the answer (Week 4).

(W6) A stochastic matrix P has entries ≥ 0 and columns summing to 1; a distribution p becomes Pp after one step (Week 3).

§2 This week

§2.1 The quotient map and the dimension of a quotient (Ch. 9.1)

Definition (Def. 25 / LADR 3.104)**.** For a subspace $U \subseteq V$, the **quotient map** is $\pi\colon V \to V/U$, $\pi(v) = v + U$.

- π is linear:
$$\pi(\lambda v + w) = (\lambda v + w) + U = \lambda(v + U) + (w + U)$$
by the definition of the operations (W5).
- π is surjective: every element of V/U is $v + U = \pi(v)$ for some v.
- $\operatorname{null} \pi = U$: $\pi(v)$ is the zero element $0 + U$ exactly when $v - 0 \in U$ (W5).

Proposition (Prop. 40 / LADR 3.105: dimension of a quotient)
If V is finite-dimensional, $\dim(V/U) = \dim V - \dim U$.

Proof. Apply rank–nullity (W2) to π. Its null space is U and its range is all of V/U. So $\dim V = \dim U + \dim(V/U)$. □

- **A basis of V/U.** Let $u_1, \ldots, u_k$ be a basis of U and extend it to a basis $u_1, \ldots, u_k, x_1, \ldots, x_r$ of V. Then $x_1 + U, \ldots, x_r + U$ is a basis of V/U. Reason: π sends the basis of V to a spanning list of V/U; the u_i go to zero, so $\pi(x_1), \ldots, \pi(x_r)$ already span; and there are $r = \dim V - \dim U = \dim(V/U)$ of them.
- If V is infinite-dimensional, V/U can be finite- or infinite-dimensional.

§2.2 The first isomorphism theorem (Ch. 9.1)

The question. A linear map $T\colon V \to W$ can fail to be an isomorphism in two ways: it sends different vectors to the same place, and it misses part of W. Can both failures be removed by changing the domain and the target, without changing what T does?

Theorem (Prop. 41 / LADR 3.107: first isomorphism theorem)
Let $T \in \mathcal{L}(V, W)$. The formula

$$\widetilde{T}\colon V/\operatorname{null} T \to \operatorname{range} T, \qquad \widetilde{T}(v + \operatorname{null} T) = Tv$$

defines an isomorphism. Moreover $T = \iota \circ \widetilde{T} \circ \pi$, where $\pi\colon V \to V/\operatorname{null} T$ is the quotient map and $\iota\colon \operatorname{range} T \to W$, $\iota(w) = w$, is the inclusion.

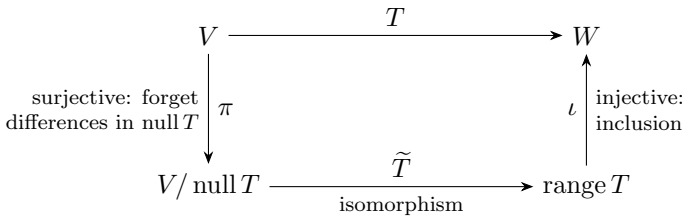
The idea

Surjectivity is repaired by shrinking the target. Replace W by $\operatorname{range} T$; then everything in the target is reached.

Injectivity is repaired by the quotient. Two vectors have the same image exactly when their difference is in $\operatorname{null} T$: $Tv = Tw \iff T(v-w) = 0 \iff v - w \in \operatorname{null} T$. And $v - w \in \operatorname{null} T$ is exactly the condition for v and w to give the same element of $V/\operatorname{null} T$ (W5). So the vectors that T cannot tell apart are precisely the ones that the quotient declares equal. After passing to the quotient there is nothing left to confuse.

That single chain of equivalences is the whole proof. Read left to right it says the formula $\widetilde{T}(v + \operatorname{null} T) = Tv$ is well-defined (same set $\Rightarrow$ same value). Read right to left it says $\widetilde{T}$ is injective (same value $\Rightarrow$ same set).

Picture: V is cut into the parallel sets $v + \operatorname{null} T$. T is constant on each set, and takes different values on different sets. So "sets $\leftrightarrow$ values" is a bijection.



Every linear map is: a quotient map (surjective), then an isomorphism, then an inclusion (injective). Going down, across and up gives the same result as T.

Proof. 1. *Well-defined.* Suppose $v + \operatorname{null} T = w + \operatorname{null} T$. Then $v - w \in \operatorname{null} T$ (W5), so $Tv - Tw = T(v-w) = 0$, so $Tv = Tw$. The value $\widetilde{T}(v + \operatorname{null} T)$ does not depend on the name v.

2. *Linear.*
$$\widetilde{T}\big(\lambda(v + \operatorname{null} T) + (w + \operatorname{null} T)\big) = \widetilde{T}\big((\lambda v + w) + \operatorname{null} T\big) = T(\lambda v + w) = \lambda Tv + Tw.$$

3. *Injective.* Suppose $\widetilde{T}(v + \operatorname{null} T) = 0$. Then $Tv = 0$, so $v \in \operatorname{null} T$, so $v + \operatorname{null} T = 0 + \operatorname{null} T$ is the zero element of the quotient.

4. *Surjective onto* $\operatorname{range} T$. Every element of $\operatorname{range} T$ is $Tv = \widetilde{T}(v + \operatorname{null} T)$ for some v.

5. *Factorisation.*
$$\iota(\widetilde{T}(\pi(v))) = \iota(\widetilde{T}(v + \operatorname{null} T)) = \iota(Tv) = Tv.$$

□

- Dimensions:
$$\dim V - \dim \operatorname{null} T = \dim(V/\operatorname{null} T) = \dim \operatorname{range} T.$$
This is rank–nullity again (not a new proof, since Prop. 40 used rank–nullity).

- Ex. 43: differentiation $D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is surjective with null space the constants $\mathbb{R}$. The induced isomorphism is $\mathcal{P}(\mathbb{R})/\mathbb{R} \to \mathcal{P}(\mathbb{R})$, $p + \mathbb{R} \mapsto p'$. Its inverse is the indefinite integral $q \mapsto \int q\,dx$, whose value is a polynomial "$+C$", i.e. an element of $\mathcal{P}(\mathbb{R})/\mathbb{R}$.
- Remark from the notes: a subspace U comes with an injective map (its inclusion), and every injective map $i\colon U \to V$ identifies U with the subspace $i(U)$. In the same way a quotient V/U comes with a surjective map π, and every surjective $T\colon V \to W$ identifies W with the quotient $V/\operatorname{null} T$.

§2.3 Induced maps; second and third isomorphism theorems (Ch. 9.2–9.3)

Proposition (Prop. 42: when a map passes to quotients)

Let $T\colon V \to W$ be linear, $U_1 \subseteq V$ and $U_2 \subseteq W$ subspaces with $T(U_1) \subseteq U_2$. Then $\widetilde{T}\colon V/U_1 \to W/U_2$, $\widetilde{T}(v + U_1) = Tv + U_2$, is a well-defined linear map.

The idea

The formula uses the name v, so the only thing to prove is that renaming does not matter. Two names v, w of the same set differ by an element of U_1. Their images Tv, Tw then differ by an element of $T(U_1)$. For the outputs to be the same set we need that difference to be in U_2. So the hypothesis $T(U_1) \subseteq U_2$ is exactly what is required, no more and no less.

Proof. 1. Suppose $v + U_1 = w + U_1$. Then $v - w \in U_1$, so $Tv - Tw = T(v - w) \in T(U_1) \subseteq U_2$, so $Tv + U_2 = Tw + U_2$.

2. Linearity:
$$\widetilde{T}(\lambda(v + U_1) + (w + U_1)) = T(\lambda v + w) + U_2 = \lambda(Tv + U_2) + (Tw + U_2).$$

□

- Special case $U_2 = \{0\}$ (HW5 P2): if $U \subseteq \operatorname{null} T$, then T induces $\widetilde{T}\colon V/U \to W$, $\widetilde{T}(v + U) = Tv$. The first isomorphism theorem is the case $U = \operatorname{null} T$.

Theorem (Prop. 43: second isomorphism theorem)

For subspaces $U, W \subseteq V$ there is an isomorphism
$$\varphi\colon U/(U \cap W) \to (U + W)/W, \qquad \varphi\big(u + (U \cap W)\big) = u + W.$$

The idea

The plan for every statement "quotient ≅ something" is the same: find a surjective linear map from the numerator onto the something, compute its null space, and apply the first isomorphism theorem. Here the numerator is U and the target is $(U + W)/W$, and the only reasonable map is $u \mapsto u + W$.

In finite dimensions the theorem gives $\dim U - \dim(U \cap W) = \dim(U + W) - \dim W$, which is the formula for $\dim(U + W)$ from Week 2.

Proof. 1. Define $S\colon U \to (U + W)/W$ by $S(u) = u + W$. It is linear (it is the quotient map of $U + W$ restricted to U).

2. *S is surjective.* An element of $(U + W)/W$ is $(u + w) + W$ with $u \in U$, $w \in W$. Since $(u + w) - u = w \in W$, this equals $u + W = S(u)$.

3. $\operatorname{null} S = U \cap W$. For $u \in U$: $S(u) = 0 + W \iff u \in W \iff u \in U \cap W$.

4. By the first isomorphism theorem, $U/\operatorname{null} S \to \operatorname{range} S$, i.e. $U/(U \cap W) \to (U + W)/W$, $u + (U \cap W) \mapsto u + W$, is an isomorphism.

□

Theorem (Prop. 44: third isomorphism theorem)
Let $W \subseteq U \subseteq V$ be subspaces. Then U/W is a subspace of V/W (the sets $u+W$ with $u \in U$), and

$$(V/W)/(U/W) \;\cong\; V/U.$$

Proof. 1. The identity map of V sends W into U. By Prop. 42 it induces $S\colon V/W \to V/U$, $S(v+W) = v+U$.

2. S is surjective: every $v+U$ is $S(v+W)$.

3. $\operatorname{null} S = U/W$: $S(v+W) = 0+U \iff v \in U \iff v+W \in U/W$.

4. The first isomorphism theorem gives $(V/W)/(U/W) \cong V/U$.

□

- Dimension check: $(\dim V - \dim W) - (\dim U - \dim W) = \dim V - \dim U$.
- "Natural" in the notes means: the isomorphism is given by a formula that uses only the subspaces themselves, with no choice of basis or complement.
- Use in induction proofs: if $W \neq \{0\}$, then V/W has smaller dimension than V.

§2.4 Dual space and dual basis (Ch. 10.1)

The question. A vector can be studied through numbers that depend linearly on it: a coordinate, the value of a polynomial at a point, an integral. What is the space of all such linear measurements, and how is it related to V?

Definition (Def. 26, 27 / LADR 3.108–3.112)**.** A **linear functional** on V is a linear map $\varphi\colon V \to \mathbb{F}$. The **dual space** is $V' = \mathcal{L}(V, \mathbb{F})$. If $v_1, \dots, v_n$ is a basis of V, its **dual basis** is the list $\varphi_1, \dots, \varphi_n \in V'$ with

$$\varphi_j(v_k) = \delta_{jk} = \begin{cases} 1 & j = k \\ 0 & j \neq k. \end{cases}$$

- V' has nothing to do with derivatives. Other books (and the practice midterm) write V^*.
- φ_j exists and is unique because a linear map is determined by its values on a basis (W1).
- **What φ_j does:** $\varphi_j(a_1 v_1 + \cdots + a_n v_n) = a_j$. It reads off the j-th coordinate.
- Ex. 44: for $\mathbb{F}^n$ (columns) a functional is a row vector, acting by row times column. The dual basis of $e_1, \dots, e_n$ is the list of rows $(1, 0, \dots, 0), \dots, (0, \dots, 0, 1)$.
- $\dim V' = \dim V \cdot \dim \mathbb{F} = \dim V$ (W4). So $V' \cong V$ when V is finite-dimensional. But an isomorphism has to be chosen (for example $v_j \mapsto \varphi_j$), and a different basis gives a different one.

Proposition (Prop. 45 / LADR 3.116: the dual basis is a basis of V')

The idea

The one technique for functionals: *to get information out of an equation between functionals, evaluate both sides on basis vectors.* Evaluating $\lambda_1\varphi_1 + \cdots + \lambda_n\varphi_n$ at v_k makes every term vanish except the k-th, and returns λ_k.

Proof. 1. $\dim V' = n$, so it is enough to prove that $\varphi_1, \dots, \varphi_n$ is independent (Week 2: n independent vectors in dimension n).

2. Suppose $\lambda_1\varphi_1 + \cdots + \lambda_n\varphi_n = 0$ (the zero functional). Evaluate at v_k:

$$0 = \lambda_1\varphi_1(v_k) + \cdots + \lambda_n\varphi_n(v_k) = \lambda_k.$$

This holds for every k.

□

- **Coordinates of a vector:** $v = \sum_j \varphi_j(v)\, v_j$. **Coordinates of a functional:** $\psi = \sum_j \psi(v_j)\, \varphi_j$ (HW5 P4). The coefficient of φ_j is found by evaluating ψ at v_j.

§2.5 The double dual (Ch. 10.2)

The question. There is no preferred isomorphism $V \to V'$. Is there one from V to $V'' = (V')'$?

Definition. For $v \in V$ let $j_V(v) \in V''$ be "evaluate at v": $j_V(v)(\varphi) = \varphi(v)$ for $\varphi \in V'$. This gives $j_V \colon V \to V''$.

- $j_V(v)$ is linear in φ: $(\lambda\varphi + \psi)(v) = \lambda\varphi(v) + \psi(v)$. So $j_V(v) \in V''$.
- j_V is linear in v: $\varphi(\lambda v + w) = \lambda\varphi(v) + \varphi(w)$.
- The definition uses no basis and no inner product.

Proposition (Prop. 46 / LADR 3F.32)

j_V is injective. If V is finite-dimensional, j_V is an isomorphism.

The idea

Injective means: if $v \neq 0$ then $j_V(v) \neq 0$, i.e. some functional φ has $\varphi(v) \neq 0$. So the task is to build a functional that does not vanish at a given nonzero vector. Functionals are built by prescribing values on a basis, so make v a basis vector and give it the value 1.

Restated: if $\varphi(v) = \varphi(w)$ for every $\varphi \in V'$, then $v = w$. A vector is determined by the results of all linear measurements.

Proof. 1. Let $v \neq 0$. The list v is independent; extend it to a basis $v, v_2, \ldots$ of V (W1).

2. Let φ be the functional with $\varphi(v) = 1$ and $\varphi(v_k) = 0$ for the other basis vectors.

3. Then $j_V(v)(\varphi) = \varphi(v) = 1 \neq 0$, so $j_V(v) \neq 0$. Hence $\operatorname{null} j_V = \{0\}$.

4. If $\dim V < \infty$: $\dim V'' = \dim V' = \dim V$, and an injective map between spaces of equal finite dimension is an isomorphism (W3).

□

In coordinates: vectors of $\mathbb{F}^n$ are columns, functionals are rows, and functionals on rows are columns again. j_V is the basis-free version of "transpose twice".

§2.6 The dual map and the transpose (Ch. 10.3)

Definition (Def. 28 / LADR 3.118)**.** For $T \in \mathcal{L}(V, W)$ the **dual map** is $T' \colon W' \to V'$, $T'(\varphi) = \varphi \circ T$.

The idea

Why does the direction reverse? A functional φ on W measures vectors of W. Given $T \colon V \to W$, we can use φ to measure a vector v of V: first send it to W, then measure. That is $\varphi \circ T$, a functional on V. So T converts measurements on W into measurements on V. There is no way to go the other way: a functional on V gives no information about vectors of W outside the range.

The defining equation to use in every computation: $(T'\varphi)(v) = \varphi(Tv)$.

- Ex. 45: $T \colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $Tp = xp$, and $\varphi(p) = p(2)$. Then $(T'\varphi)(p) = \varphi(xp) = 2p(2)$, so $T'\varphi = 2\varphi$.
- Practice Midterm 2: $T(x, y) = 2x - y$ from $\mathbb{R}^2$ to $\mathbb{R}$ and $\ell(t) = 3t$. Then $(T'\ell)(x, y) = \ell(2x - y) = 6x - 3y$.

Proposition (Prop. 47 / LADR 3.120: algebra of dual maps)
$(\lambda T_1 + T_2)' = \lambda T_1' + T_2'$, and $(ST)' = T'S'$ for $T \in \mathcal{L}(U,V)$, $S \in \mathcal{L}(V,W)$.

Proof. 1. For $\varphi \in W'$ and $v \in V$:

$$\big((\lambda T_1 + T_2)'\varphi\big)(v) = \varphi(\lambda T_1 v + T_2 v) = \lambda\varphi(T_1 v) + \varphi(T_2 v) = \big(\lambda T_1'\varphi + T_2'\varphi\big)(v).$$

2. Apply the definition of the dual map three times, using that composition is associative:

$$(ST)'\varphi = \varphi \circ (S \circ T) = (\varphi \circ S) \circ T = (S'\varphi) \circ T = T'(S'\varphi).$$

□

Proposition (Prop. 48: j is compatible with every linear map)
For $T\colon V \to W$: $T'' \circ j_V = j_W \circ T$.

Proof. For $v \in V$ and $\psi \in W'$, unwind the definitions one at a time:

$$\big(T''(j_V(v))\big)(\psi) = j_V(v)(T'\psi) = (T'\psi)(v) = \psi(Tv) = j_W(Tv)(\psi).$$

□

This is what "canonical" means for j: under the identifications $V \cong V''$, $W \cong W''$, the map T'' is T.

Theorem (Prop. 49 / LADR 3.132: the matrix of the dual map is the transpose)
Let $B = (v_1, \dots, v_n)$, $C = (w_1, \dots, w_m)$ be bases of V, W with dual bases $B' = (v_1', \dots, v_n')$, $C' = (w_1', \dots, w_m')$. If $A = \mathcal{M}(T)$ with respect to B, C, then $\mathcal{M}(T')$ with respect to C', B' is A^{T}, where $(A^{\mathsf{T}})_{ij} = a_{ji}$.

The idea

Column j of $\mathcal{M}(T')$ must hold the coordinates of the functional $T'(w_j') = w_j' \circ T$ in the basis $v_1', \dots, v_n'$. What does $w_j' \circ T$ do? It sends v to the j-th coordinate of Tv. In coordinates, the j-th coordinate of Ax is $a_{j1}x_1 + \cdots + a_{jn}x_n$: the coefficients are *row j of A*. So row j of A becomes column j of $\mathcal{M}(T')$. That is the transpose.

To prove it, find the coordinates of $T'(w_j')$ by evaluating at each v_k (the technique from Prop. 45).

Proof. 1. By definition of A: $Tv_k = \sum_{l=1}^{m} a_{lk} w_l$.

2. Evaluate $T'(w_j')$ at v_k:

$$T'(w_j')(v_k) = w_j'(Tv_k) = w_j'\Big(\sum_l a_{lk} w_l\Big) = a_{jk}.$$

3. The coordinates of a functional ψ in the dual basis are its values $\psi(v_k)$. So $T'(w_j') = \sum_{k=1}^{n} a_{jk} v_k'$.

4. So column j of $\mathcal{M}(T')$ is $(a_{j1}, \dots, a_{jn})^{\mathsf{T}}$, which is row j of A.

□

- Ex. 46: $A = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{smallmatrix}\right)\colon \mathbb{R}^3 \to \mathbb{R}^2$. The dual map sends a row $(x\ y)$ to the row $(x\ y)A$; its matrix in the standard dual bases is

$$A^{\mathsf{T}} = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

- $(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$, because $(ST)' = T'S'$.
- **Distributions and observables.** For a Markov chain with stochastic matrix P (W6), let f assign a reward f_i to state i. The mean reward under a distribution p is $f^{\mathsf{T}}p = \sum f_i p_i$, a linear functional of p. After one step it is $f^{\mathsf{T}}Pp = (P^{\mathsf{T}}f)^{\mathsf{T}}p$. So P moves distributions forward, $p \mapsto Pp$, and the dual map moves reward functions, $f \mapsto P^{\mathsf{T}}f$, with $(P^{\mathsf{T}}f)_j = \sum_i P_{ij} f_i =$ the expected reward one step later, given the present state is j. This is why some books use rows and P^{T}.

§2.7 Annihilators (Ch. 10.4)

Definition (Def. 29 / LADR 3.121)**.** The **annihilator** of a subspace $U \subseteq V$ is

$$U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for all } u \in U\}.$$

It is a subspace of V'.

- It is enough to check φ on a spanning list of U.
- Ex. 47: $U = \operatorname{span}\big((1,1,0),(0,1,1)\big)$. A row $(a\ b\ c)$ is in U^0 when $a+b=0$ and $b+c=0$, i.e. $(a\ b\ c) = b(-1\ \ 1\ \ -1)$. So $U^0 = \operatorname{span}\big((-1\ \ 1\ \ -1)\big)$.
- $\{0\}^0 = V'$ and $V^0 = \{0\}$.

Proposition (LADR 3.125: dimension of the annihilator)
If V is finite-dimensional, $\dim U^0 = \dim V - \dim U$.

Proof. 1. Let $u_1, \dots, u_k$ be a basis of U; extend to a basis $u_1, \dots, u_k, x_1, \dots, x_r$ of V, with dual basis $\varphi_1, \dots, \varphi_k, \chi_1, \dots, \chi_r$.

2. Any $\psi \in V'$ is $\psi = \sum_i \psi(u_i)\varphi_i + \sum_j \psi(x_j)\chi_j$ (coordinates of a functional).

3. $\psi \in U^0$ exactly when $\psi(u_i) = 0$ for all i, i.e. exactly when $\psi \in \operatorname{span}(\chi_1, \dots, \chi_r)$.

4. So $\chi_1, \dots, \chi_r$ is a basis of U^0, and $r = \dim V - \dim U$.

□

Theorem (Prop. 50 / LADR 3.128, 3.130: null space and range of the dual map)
For $T \in \mathcal{L}(V,W)$: (i) $\operatorname{null} T' = (\operatorname{range} T)^0$; (ii) $\operatorname{range} T' = (\operatorname{null} T)^0$.

The idea

(i) is a translation. $T'\varphi = 0$ means $\varphi(Tv) = 0$ for all v, which says φ vanishes on every vector of the form Tv, i.e. on $\operatorname{range} T$.

(ii) has an easy half and a half that needs a construction. Easy: a functional $\psi \circ T$ vanishes wherever T does. Harder: given $\varphi \in V'$ that vanishes on $\operatorname{null} T$, we must produce $\psi \in W'$ with $\psi \circ T = \varphi$. The equation tells us what ψ has to be on $\operatorname{range} T$: $\psi(Tv) = \varphi(v)$. Is that a valid definition? The same vector of $\operatorname{range} T$ can be Tv and Tw; then $v - w \in \operatorname{null} T$, and $\varphi(v) = \varphi(w)$ precisely because φ vanishes on $\operatorname{null} T$. So the hypothesis is exactly the well-definedness condition. Outside $\operatorname{range} T$ the equation imposes nothing, so extend ψ in any way.

Proof. 1. (i)

$$\varphi \in \operatorname{null} T' \iff \varphi \circ T = 0 \iff \varphi(Tv) = 0$$

for all $v \in V \iff \varphi \in (\operatorname{range} T)^0$.

2. (ii), $\subseteq$. Let $\varphi = T'\psi = \psi \circ T$. For $v \in \operatorname{null} T$: $\varphi(v) = \psi(Tv) = \psi(0) = 0$. So $\varphi \in (\operatorname{null} T)^0$.

3. (ii), $\supseteq$. Let $\varphi \in (\operatorname{null} T)^0$. Define $\psi_0 \colon \operatorname{range} T \to \mathbb{F}$ by $\psi_0(Tv) = \varphi(v)$. Well-defined: if $Tv = Tw$ then $v - w \in \operatorname{null} T$, so $\varphi(v) - \varphi(w) = \varphi(v - w) = 0$. It is linear.

4. Extend a basis of $\operatorname{range} T$ to a basis of W, and let $\psi \in W'$ equal ψ_0 on the basis vectors of $\operatorname{range} T$ and 0 on the added ones. Then $\psi = \psi_0$ on $\operatorname{range} T$.

5. $T'\psi = \psi \circ T$ sends v to $\psi(Tv) = \psi_0(Tv) = \varphi(v)$. So $T'\psi = \varphi$ and $\varphi \in \operatorname{range} T'$.

□

Proposition (Prop. 51 / LADR 3.129, 3.131)
T is injective $\iff$ T' is surjective. T is surjective $\iff$ T' is injective.

Proof. 1. T injective

$$\iff \operatorname{null} T = \{0\} \iff (\operatorname{null} T)^0 = V' \iff \operatorname{range} T' = V'.$$

(For the middle step in the direction $\Leftarrow$: if $\operatorname{null} T$ contained some $v \neq 0$, the functional built in Prop. 46 has $\varphi(v) = 1$, so it is not in $(\operatorname{null} T)^0$.)

2. T surjective

$$\iff \operatorname{range} T = W \iff (\operatorname{range} T)^0 = \{0\} \iff \operatorname{null} T' = \{0\}.$$

(Middle step, $\Leftarrow$: if $\operatorname{range} T \neq W$, extend a basis of $\operatorname{range} T$ by at least one vector x; the functional that is 1 on x and 0 on the other basis vectors is a nonzero element of $(\operatorname{range} T)^0$.)

□

- $\dim \operatorname{range} T' = \dim \operatorname{range} T$ (finite dimensions):

$$\dim \operatorname{range} T' = \dim (\operatorname{null} T)^0 = \dim V - \dim \operatorname{null} T = \dim \operatorname{range} T.$$

For a matrix: A and A^T have the same rank, i.e. the span of the rows and the span of the columns have the same dimension.

Shortcuts

- **Proving "quotient $\cong X$"**: write a surjective linear map from the numerator onto X, find its null space, quote the first isomorphism theorem. No need to check well-definedness by hand.
- **Defining a map out of V/U**: the formula $v + U \mapsto Tv$ is valid exactly when $U \subseteq \operatorname{null} T$.
- **Basis of V/U**: extend a basis of U to a basis of V; the added vectors $+U$ are a basis. $\dim V/U = \dim V - \dim U$.
- **Basis of $V/\operatorname{null} T$ when T is onto $\mathbb{F}^k$**: pick $p_1, \ldots, p_k$ with $Tp_i = e_i$; then $p_i + \operatorname{null} T$ is a basis (HW5 P1, Practice Midterm 2 Q1).
- **Equation between functionals**: evaluate on a basis. The j-th coordinate of ψ in the dual basis is $\psi(v_j)$.
- **Dual basis in $\mathbb{F}^n$**: put the basis vectors as the columns of M; the dual basis is the rows of M^{-1} (because $M^{-1}M = I$ says row j times column k is δ_{jk}).
- **A nonzero functional is surjective**, so its null space has dimension $\dim V - 1$ (HW5 P3).
- **Matrix of T'**: transpose. $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$. $\operatorname{rank} A^\mathsf{T} = \operatorname{rank} A$.
- **Annihilator in $\mathbb{F}^n$**: rows $(a_1 \ \ldots \ a_n)$ with $(\text{row}) \cdot u = 0$ for each spanning vector u of U; a linear system. Expect dimension $n - \dim U$.
- **Injective/surjective swap under the dual**; $\operatorname{null} T' = (\operatorname{range} T)^0$, $\operatorname{range} T' = (\operatorname{null} T)^0$.

Traps

- Elements of V/U are sets. A basis of V/U consists of elements like $x + U$, not vectors x.
- The dual basis depends on the whole basis: changing v_2 can change φ_1.
- $V \cong V'$ in finite dimensions, but not by a preferred map; $V \cong V''$ by the preferred map j_V. In infinite dimensions j_V is injective but need not be surjective.
- T' goes from W' to V', and $(ST)' = T'S'$: orders reverse.
- U^0 is a subspace of V', not of V.
- V' is not a derivative; T' is not the derivative of T.

§3 Problem set

HW5 P1. Define $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathbb{R}^2$, $T(p) = (p(0), p(1))$. (a) Determine $\operatorname{null} T$ and prove that T is surjective. (b) Describe explicitly the induced isomorphism $\mathcal{P}_3(\mathbb{R})/\operatorname{null} T \to \mathbb{R}^2$ from the first isomorphism theorem. (c) Find a basis of $\mathcal{P}_3(\mathbb{R})/\operatorname{null} T$.

Hint: a polynomial with zeros at 0 and 1 is divisible by $x(x-1)$. For (c), an isomorphism carries a basis back to a basis.

HW5 P2. Let $T\colon V \to W$ be linear and U a subspace of V contained in $\operatorname{null} T$. Prove that there is a unique linear map $\widetilde{T}\colon V/U \to W$ with $\widetilde{T}(v+U) = Tv$ for every v. Determine $\operatorname{null} \widetilde{T}$ in terms of $\operatorname{null} T$ and U.

Hint: well-definedness is the only issue for existence.

HW5 P3 (LADR 3F, Exercise 1). Prove that every linear functional $\varphi\colon V \to \mathbb{F}$ is either the zero map or surjective.

Hint: $\operatorname{range}\varphi$ is a subspace of the one-dimensional space $\mathbb{F}$; or scale a vector with $\varphi(v) \neq 0$.

HW5 P4. Let $v_1, \dots, v_n$ be a basis of V with dual basis $\varphi_1, \dots, \varphi_n$. (a) Prove $v = \sum_{j=1}^{n} \varphi_j(v) v_j$ for every $v \in V$. (b) Prove $\psi = \sum_{j=1}^{n} \psi(v_j)\varphi_j$ for every $\psi \in V'$.

Hint: (a) apply φ_j to $v = \sum a_k v_k$. (b) evaluate both sides at v_k.

HW5 P5. Let $T\colon V \to W$ be linear between finite-dimensional spaces with matrix A in bases B, C. Prove that the matrix of $T'\colon W' \to V'$, $T'(\psi) = \psi \circ T$, in the dual bases is A^{T}.

Hint: evaluate $T'(w_j')$ at v_k.

HW5 P6. Let $U = \operatorname{span}\big((1,1,0),(0,1,1)\big) \subseteq \mathbb{R}^3$. (a) Determine $U^0 \subseteq (\mathbb{R}^3)'$, writing functionals as row vectors. (b) Verify that $\dim U + \dim U^0 = 3$.

HW5 P7 (optional). Let $j_V\colon V \to V''$, $j_V(v)(\varphi) = \varphi(v)$. Prove that j_V is injective, and an isomorphism when V is finite-dimensional.

R1 (Practice Midterm 2, Question 1). Let $V = \mathcal{P}_3(\mathbb{R})$, $E\colon V \to \mathbb{R}^2$, $E(p) = (p(0), p(1))$, $U = \operatorname{null} E$. (a) Prove that E is linear and surjective. (b) Find a basis of U. (c) Prove that $\widetilde{E}(p+U) = E(p)$ defines a linear isomorphism $V/U \to \mathbb{R}^2$. (d) Give a basis of V/U and the images of its elements under $\widetilde{E}$.

R2 (Practice Midterm 2, Question 2). Let $T\colon V \to W$, $S\colon W \to Z$ be linear maps between finite-dimensional real spaces, with dual maps $T^*\colon W^* \to V^*$, $S^*\colon Z^* \to W^*$. (a) Give the definition of T^* and prove it is linear. (b) Prove $(S \circ T)^* = T^* \circ S^*$. (c) Prove that if T is surjective then T^* is injective. (d) For $T\colon \mathbb{R}^2 \to \mathbb{R}$, $T(x,y) = 2x - y$ and $\ell(t) = 3t$, compute $T^*\ell$.

E1 (dual basis computation). Find the dual basis of the basis $v_1 = (1,1)$, $v_2 = (1,-1)$ of $\mathbb{R}^2$, as row vectors. Then write the functional $\psi(x,y) = 3x + y$ in that dual basis.